

WEEK 3: DASHBOARD DEVELOPMENT AND INSIGHTS REPORT

1. Dashboard Overview

This deliverable presents an interactive Power BI dashboard built on a cleaned dataset of 7,151 DePaul University graduate applicants, covering recruitment sources, admissions outcomes, and counselor engagement activity. The dashboard is organized into three linked pages, each targeting one of the focus areas required for this deliverable: Recruitment Overview, Admissions Funnel, and Counselor Engagement. The accompanying .pbix file and/or published Power BI link is submitted alongside this report.

2. Dashboard Design Documentation

2.1 Layout and Structure

The dashboard uses a consistent three-page structure so that a reader can move from a high-level recruitment picture, into the admissions process, and finally into the counselor activity that drives outcomes. Each page follows the same visual grammar: KPI cards across the top, a row of slicers for filtering, and supporting charts beneath, so that the dashboard feels familiar regardless of which page is open.

- **Page 1 — Recruitment Overview:** Provides a quick top-line summary (Total Applicants, Total Admitted, Admit Rate) before any detailed exploration.
- **Page 2 — Admissions Funnel:** Walks the applicant through the admissions process stage by stage and surfaces where cases stall.
- **Page 3 — Counselor Engagement:** Isolates counselor activity and ties it directly back to admit outcomes.

2.2 Visualization Choices and Rationale

Each visual was chosen to match the shape of the question it answers, rather than defaulting to one chart type across the dashboard:

Visual	Page	Why it was included
KPI Cards	All pages	Give an instant, filter-aware snapshot of the most important number on each page

		(volume, rate, or workload) without requiring the reader to interpret a chart.
Map (bubble)	Recruitment	Country is a geographic attribute; a map shows applicant concentration at a glance far better than a table of 95 countries would.
Bar / Column charts	All pages	Best for ranking categories with many labels (majors, callers, outcomes) — long labels read more easily on a horizontal bar.
Donut charts	Recruitment	Used only where a field has a small number of categories (Application Source; Intake Degree Type), so the part-to-whole split stays readable.
Line chart	Recruitment	Shows the Year/Season hierarchy as a trend, making seasonal intake patterns (e.g. Fall peaks) visible over time.
Funnel chart	Admissions	Purpose-built for sequential, drop-off processes — makes stage-by-stage attrition in the admissions workflow immediately visible.
Stacked bar	Admissions, Counselor	Shows both a total and its composition in one visual (e.g. degree type by admit outcome; case status mix per counselor).
Sortable table	Admissions	Lets a reader drill into exact figures by major and re-sort by admit rate — detail a chart alone cannot provide.
Slicers (dropdown)	All pages	Continent, Year, Season, Application Source, Degree Type, Country, Status, and Caller slicers let any user self-serve a question without editing the report.

2.3 Data Flow and Interactivity Features

- All visuals on a page read from a single underlying table (Applicant_Data) loaded once into Power BI, so every slicer and click filters every visual on that page consistently — no separate queries to keep in sync.
- Selecting a value in any chart (e.g. a country bubble on the map) cross-filters every other visual on the page automatically; this is Power BI's native cross-filtering and requires no extra configuration.
- Dropdown slicers (Continent, Year, Season, Application Source, Degree Type, Country, Status, Caller) give multi-level drill-down without needing separate pages for each segment.
- Hovering over any bar, bubble, or chart segment surfaces an exact value tooltip, so the dashboard stays uncluttered while still exposing precise figures on demand.
- Nine DAX measures (Total Applicants, Total Admitted, Admit Rate, Total Engaged, Engagement Rate, Avg Attempts per Applicant, Avg Days Contact to Admit, and Admit Rate split by Engaged/Not Engaged) recalculate live as filters change, so every KPI card always reflects the current selection.

3. Insight Report

3.1 Key Trends and Patterns

- **Overall funnel:** Of 7,151 applicants in the dataset, 5,398 were admitted, for an overall admit rate of 75.5%.
- **Recruitment is geographically concentrated:** India accounts for 4,348 applicants (61% of the dataset), more than five times the next-largest source, the United States (709). Ghana, Nigeria, and Pakistan round out the top five — recruitment is heavily India-concentrated.
- **Program demand is concentrated in a few programs:** Business Analytics (1,017) and Computer Science (985) are the two largest intake majors, together accounting for over a quarter of all applicants; the MS degree type alone makes up 81% of intake.
- **Application source split:** 86% of applicants come through the Non-Study Group channel versus 14% through Study Group — Study Group is a small but identifiable acquisition channel worth tracking separately for ROI.
- **Counselor engagement is strongly associated with admission:** This is the most striking pattern in the data: applicants a counselor has actually engaged with (called,

emailed, or contacted) are admitted 98.8% of the time, versus 46.6% for applicants with no recorded engagement — a 52-point gap.

- **Nearly half of all applicants have not yet been engaged:** 3,194 applicants (45% of the dataset) show no counselor activity at all (“Not Yet Assigned” / “No Activity Logged”), representing the single largest pool of un-actioned leads in the data.
- **Engaged-case performance is consistently high across counselors:** Among engaged applicants, the busiest counselor (Jagdeep, 1,940 cases) and the others (Swarna, Ayesha, Lavnaya) all show admit rates between 97.7% and 100% on their engaged caseload — engagement quality looks consistent across counselors, but workload is very unevenly distributed.
- **Transcript and follow-up issues are the leading process blockers:** “Follow up” (1,199 cases) and “Missing 0 Transcript” (1,126 cases) are the two largest logged blockers after the “no outcome logged” bucket, pointing to document collection and follow-up cadence as the main friction points in the process.

3.2 Actionable Recommendations

Recruitment Targeting

- Continue to invest in India-focused recruitment given its outsized share of volume, but treat the next tier (US, Ghana, Nigeria, Pakistan) as diversification targets to reduce single-country concentration risk.
- Direct marketing spends toward Business Analytics, Computer Science, and Data Science given their applicant volume, while monitoring smaller but high-admit-rate majors as efficient niches.
- Evaluate the Study Group channel’s cost-per-applicant against its 14% volume share to decide whether to scale or rebalance acquisition spend toward Non-Study Group channels.

Counselor Engagement

- Prioritize assigning a counselor to the 3,194 “not yet engaged” applicants first — this is the largest single lever available, since engaged applicants are admitted at more than double the rate of unengaged ones.
- Rebalance caseloads: Jagdeep currently carries far more volume than Swarna, Ayesha, or Lavnaya; redistributing leads more evenly could reduce response-time bottlenecks without affecting admit-rate quality, which is already consistent across counselors.

- Since 1 attempt on average is enough to engage an applicant successfully, focus coaching on getting a first attempt logged sooner for unengaged leads rather than on increasing attempts per applicant.

Admission Outcomes

- Target the “Missing Transcript” and “Follow up” backlogs directly — together they represent over 2,300 cases stalled on a specific, fixable action rather than a true decline.
- Set a service-level target (e.g. first contact within X days of application) to convert more of the “No Activity Logged” pool into the high-converting “Engaged” category before intake deadlines pass.
- Use the Admit Rate by Major table to identify low-volume, high-admit-rate programs that may be under-marketed relative to their conversion strength.

3.3 How These Insights Support Data-Driven Decision-Making

Rather than relying on anecdotal impressions of “which counselor is busy” or “where applicants come from,” the dashboard quantifies each of these questions and keeps them filterable by year, season, country, and degree type. This means admissions leadership can re-run the same analysis for any future intake cycle without rebuilding it, test the engagement-rate hypothesis on new cohorts as they arrive, and reallocate counselor workload or recruitment spend based on measured admit-rate impact rather than assumption. The 52-point admit-rate gap between engaged and unengaged applicants, in particular, gives a concrete, defensible business case for prioritizing counselor outreach as a resourcing decision.

4. Dashboard Screenshots

The three pages of the live, interactive Power BI dashboard are shown below. In the live file, all charts, maps, and tables respond to the slicers shown at the top of each page, and hovering over any data point reveals an exact-value tooltip.

Page 1 — Recruitment Overview

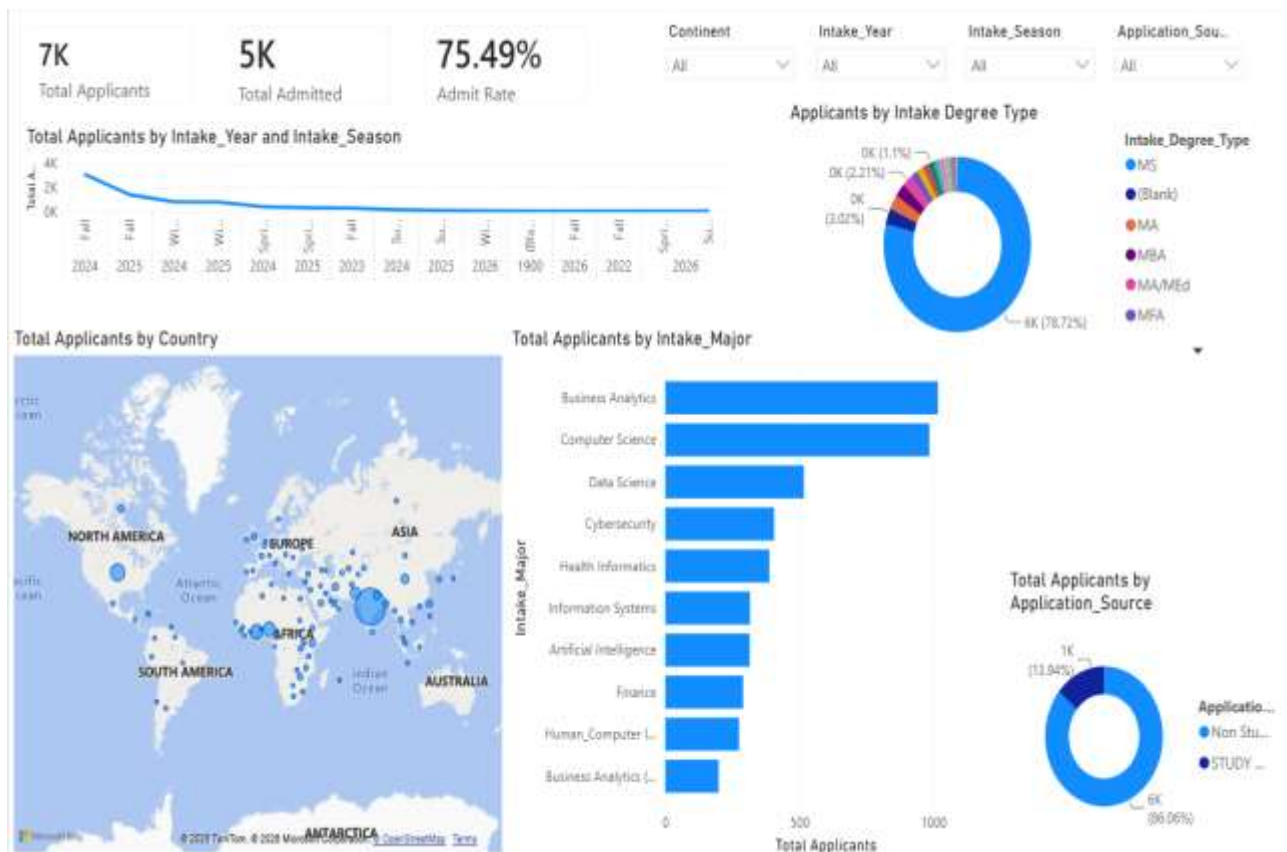


Fig 1: KPI summary, year/season trend, country map, top intake majors, application source, and intake degree type.

Page 2 — Admissions Funnel

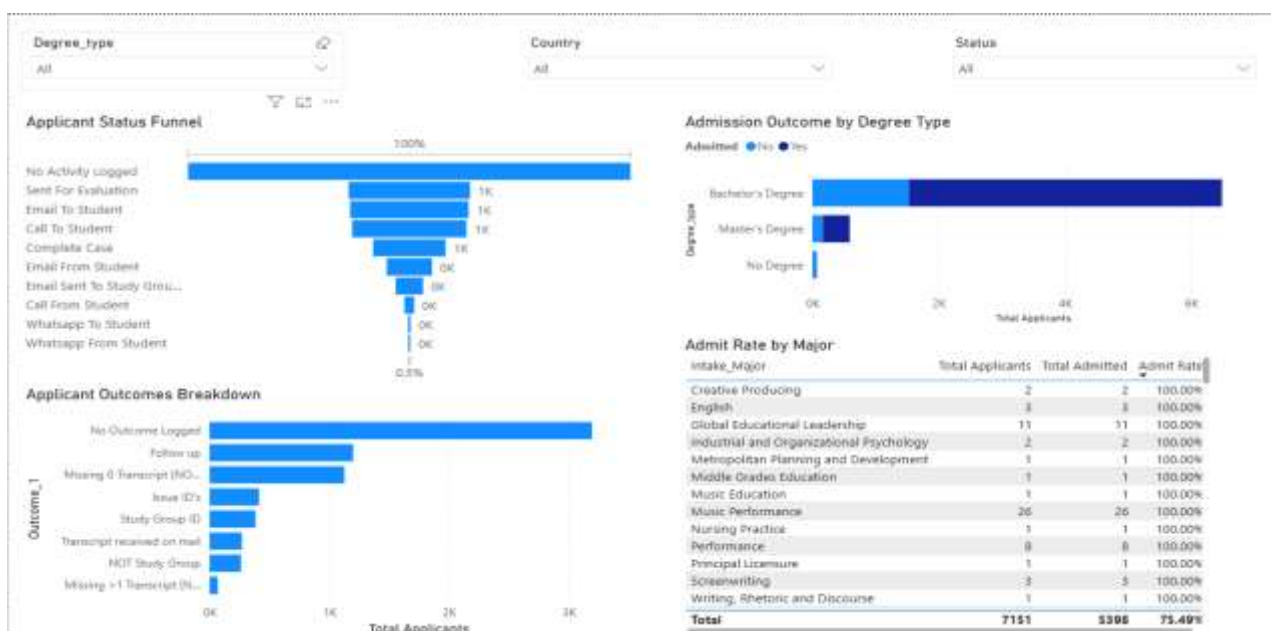


Fig 2: Applicant status funnel, admission outcome by degree type, outcomes breakdown, and admit rate by major (sortable table).

Page 3 — Counselor Engagement



Fig 3: Engagement KPIs, counselor workload vs. admits, and admit rate by counselor on engaged cases.

5. Tools and References

- Microsoft Power BI Desktop — used to build the interactive dashboard (data model, DAX measures, and report pages).
- Source dataset: DePaul University graduate applicant records (cleaned and enriched for analysis: standardized dates, derived Admitted / Engaged_By_Counselor / Enrolled_Started flags, and a Continent lookup).
- Microsoft Power BI documentation (powerbi.microsoft.com) — referenced for visual configuration and DAX syntax.

Link to Dashboard:

<https://app.powerbi.com/view?r=eyJrIjoiaGGMwNzUwZTA0OGE1MS00ZTkYLVl1ZDltN2UwYWY3YjViODA3IiwidCI6IjgzMDQ2NGFmLTgwOWEtNDgxYS1iMjE1LTgyNTU5ZWZmYWY1ZmIj9>